Artificial Intelligence II: Deep learning methods

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Lecture 5: Recurrent Neural Networks

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Overview

1. Coping with sequential data
2. *Memory Cells* in Neural Networks
3. Sequence Analysis
4. Attention in Neural Networks

Coping with sequential data

Sequential Data — Why Order Matters

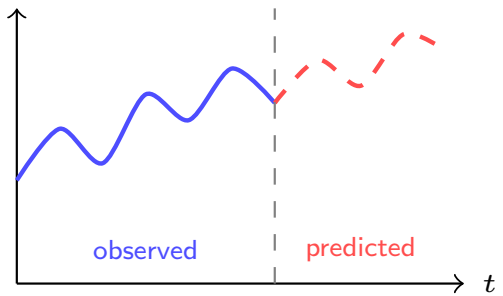
Key Insight

The meaning of each element depends on what came **before** it.

Language Modeling

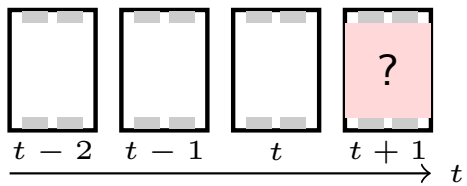
```
>The dog bit the man.
>The man bit the dog.
```

Time Series Forecasting



Today's value depends on yesterday's trend

Audio / Video Modeling



Why feedforward networks fall short:

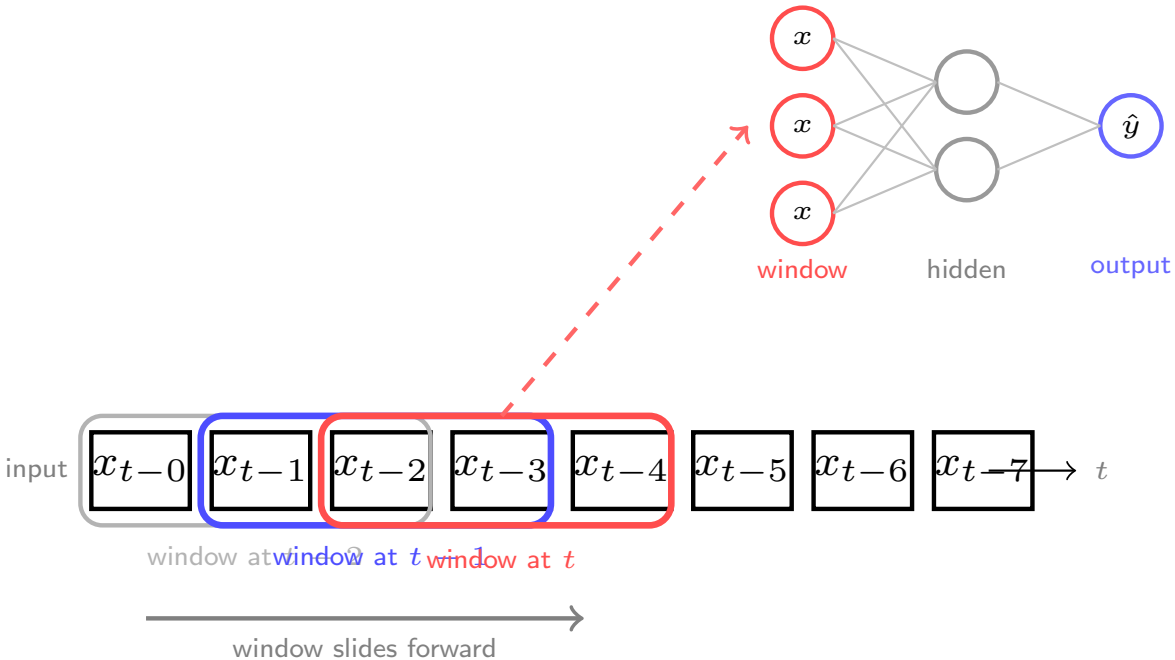
Fixed input size
Can't handle variable-length sequences

No memory
Each input processed independently

No temporal awareness
Position 1 and position 10 are unrelated features

Spatialising the time: Time-Delay Neural Networks (TDNN)

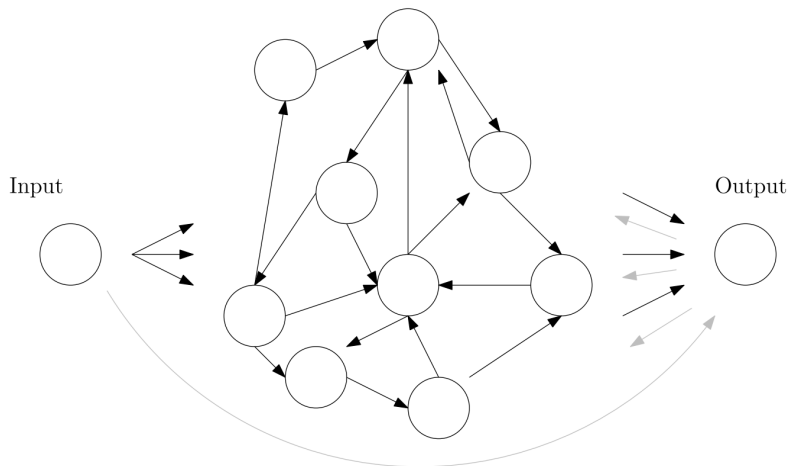
- The idea was introduced by Waibel et. al. in 1989 for phoneme recognition.



Open questions

- What size to use for the time window?
- Should the history size be constant?
- Do we need the data over the whole time span?
- How to share computation steps in time?

Recurrent Neural Network



Different types of weight matrices:

- W^{in} – input to hidden
- W^{out} – hidden to output
- W^{h} – hidden to hidden (applied multiple times)
- W^{back} – outputs to hidden

Different versions of RNNS

- if $W^{back} = 0 \rightarrow$ Elman networks
- if $W^{h} = 0 \rightarrow$ Jordan Networks
- if W^{h} is fixed $\rightarrow$ Echo state networks

Task Classification: *"M-to-N"*

1. **One to One**
   - Example: Simple Classification
   - Solution: Vanilla RNN
2. **One to Many**
   - Example: Image Captioning
   - Solution: Show and Tell
3. **Many to One**
   - Example: Sentiment Analysis
   - Solution: LSTM for Sentiment Classification
4. **Many to Many**
   - Aligned sequences of identical sizes
     - Example: Video Classification
     - Solution: Time-Distributed CNN with RNN
   - Unaligned sequences of different sizes
     - Example: Machine Translation, Automatic Speech Recognition
     - Solution: Seq2Seq Models, Listen, Attend and Spell

Different paradigms

Idea : Inputs and outputs can have variable size

Many to one

language models, sentiment analysis: multi class sequence classification

[‘this‘, ‘movie‘, ‘was‘, ‘awesome‘] $\mapsto$ 1
[‘you‘, ‘should‘, ‘was‘, ‘so‘ ‘bad‘ ‘it’s‘ ‘not‘ ‘worth‘ ‘the‘ ‘money‘] $\mapsto$ 0

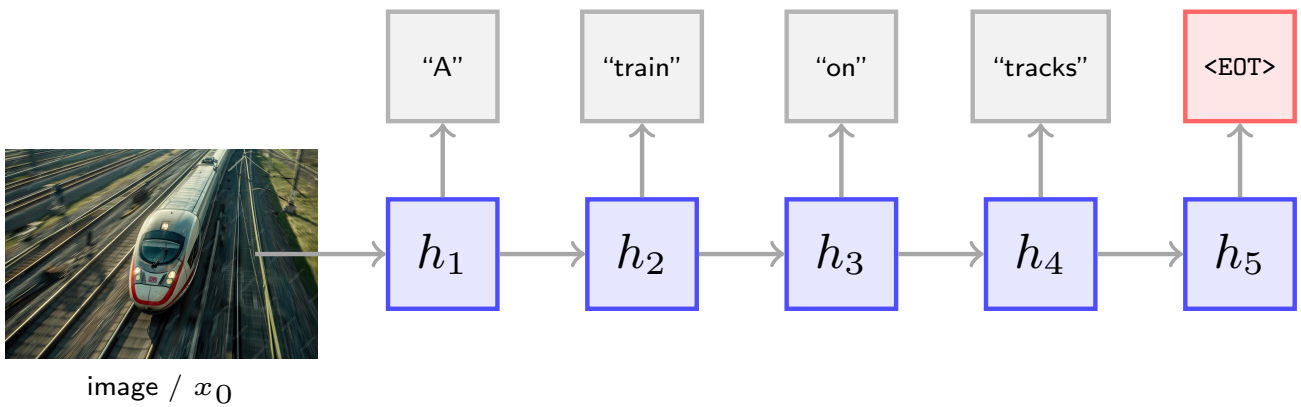
Many to many

Neural machine Translation, e.g. RO $\rightarrow$ EN
“*Războiul afectează economia globală* ” $\mapsto$ “The war is affecting the global economy”

Different Paradigms

One-to-Many — Image Captioning

A single input is processed by the network, which then generates a variable-length sequence token by token until <EOT>.



Training RNNs

Problems

- Unrolled over time, RNNs behave as **very deep** networks
- Reverse-mode differentiation is **computationally expensive**
- Prone to **exploding** or **vanishing** gradients

Some Solutions

- Forward-mode differentiation on the unrolled graph
- Truncated backprop — work on **small windows** to reduce overhead
- Gated architectures: **LSTM**, **GRU**, etc.

Takeaway

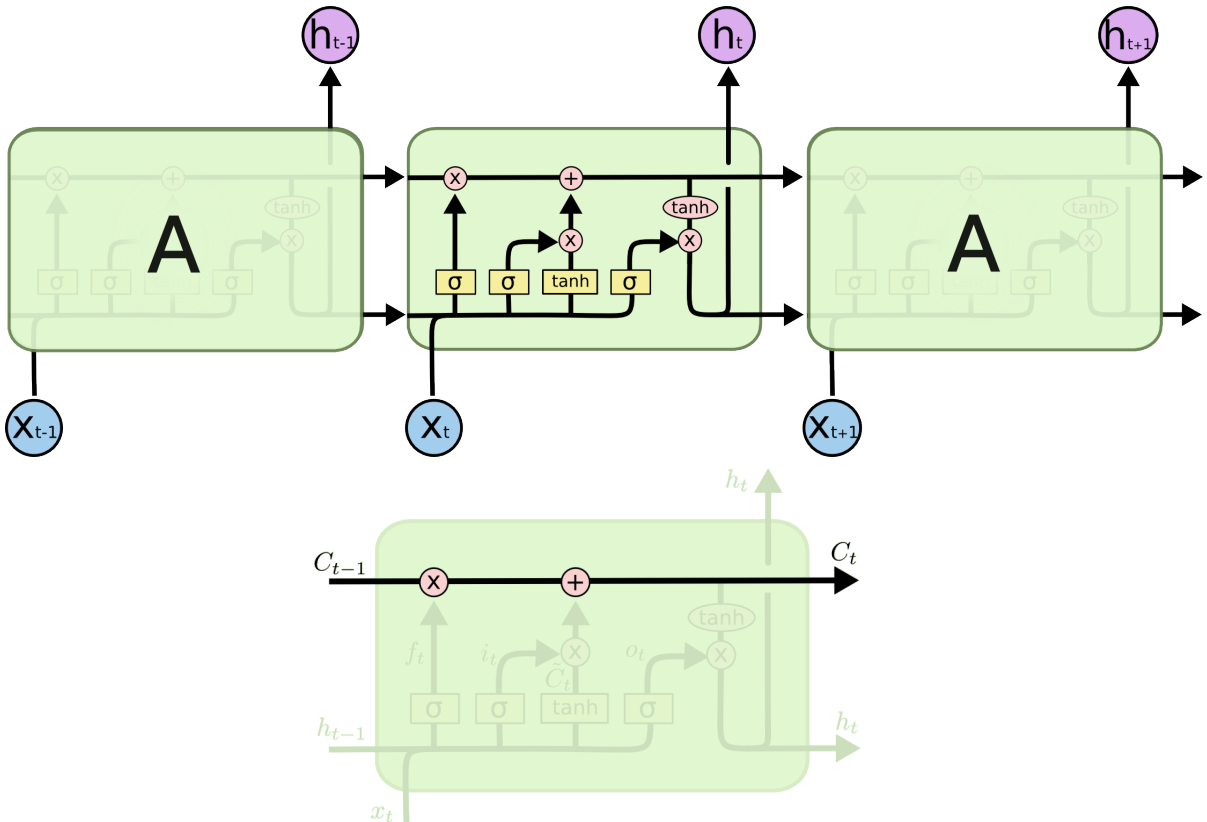
Most practical improvements trade off **exactness** of gradients for **computational feasibility** or **architectural inductive bias**.

Memory Cells in Neural Networks

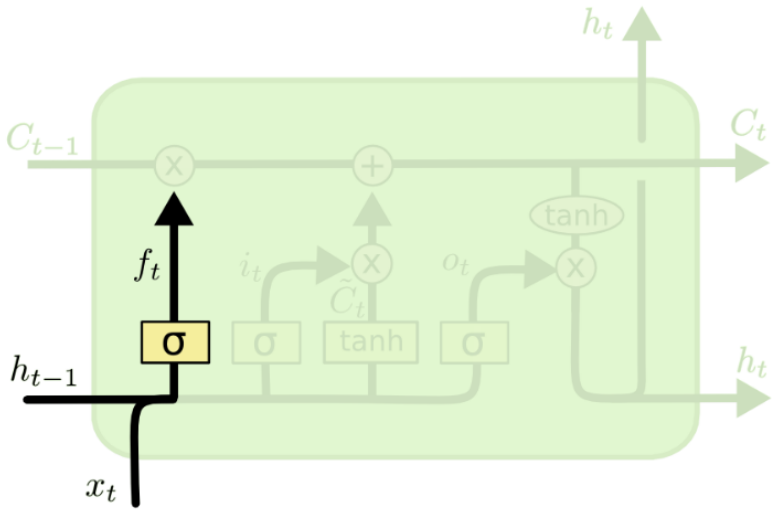
LSTM

Problem : RNNs have problems learning long range dependencies.

LSTM → Long-Short Term Memory (introduced by Schmidhuber în 1997) use memory cells to address this problem.



LSTM – step by step



$$f_t = \sigma(W_f^x x_t + W_f^h h_{t-1} + b_f) \quad \in [0, 1]$$

→ forget gate

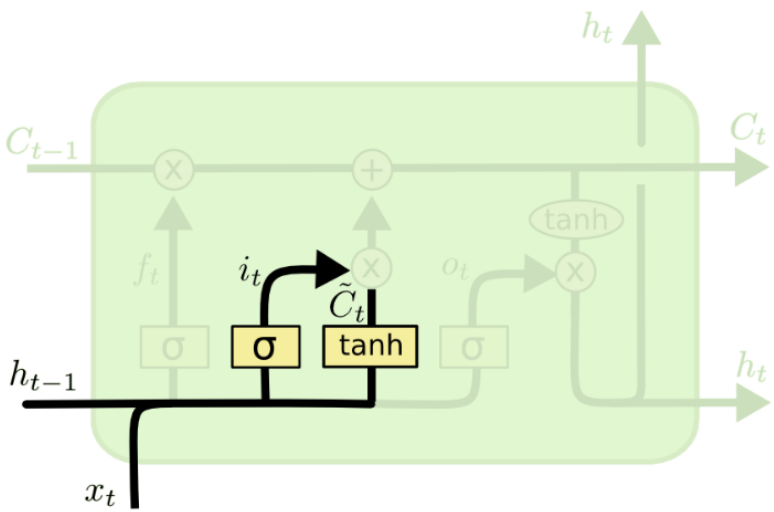
$$i_t = \sigma(W_i^x x_t + W_i^h h_{t-1} + b_i) \quad \in [0, 1]$$

→ information gate

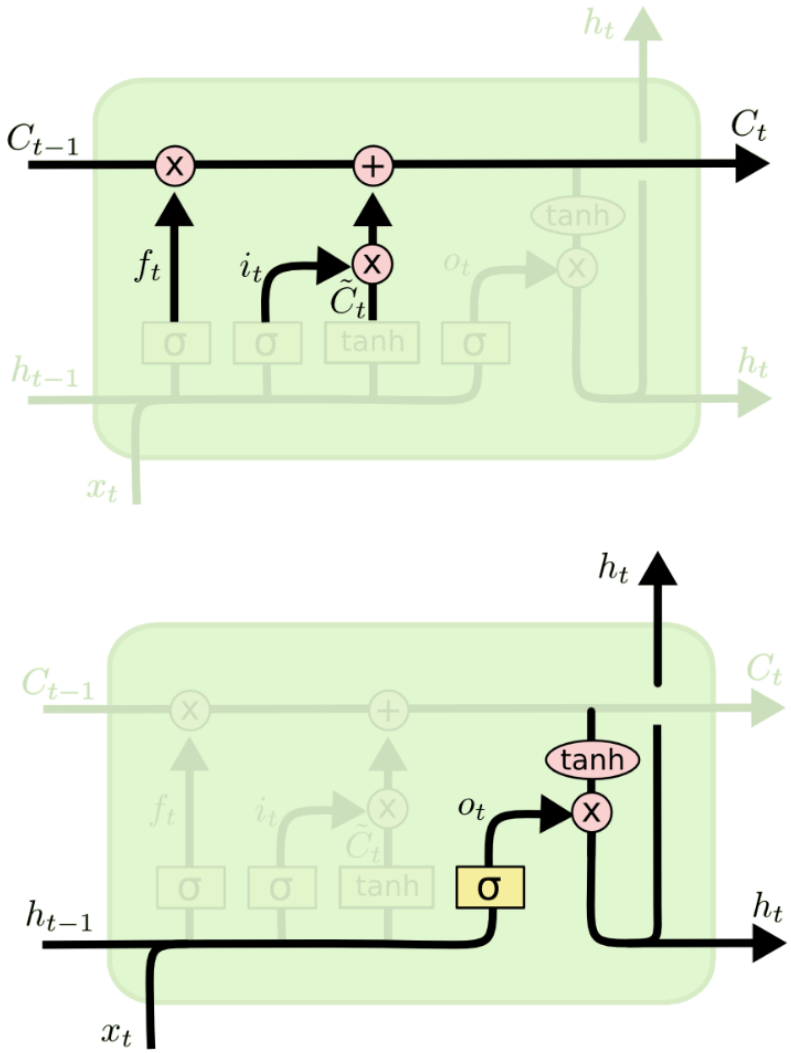
$$\tilde{C}_t = \tanh(W_C^x x_t + W_C^h h_{t-1} + b_C) \quad \in [-1, 1]$$

→ information features

$$\boxed{C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t}$$



LSTM – step by step



$$o_t = \sigma(W_o^x x_t + W_o^h h_{t-1} + b_o) \quad \in [0, 1]$$

→ output gate

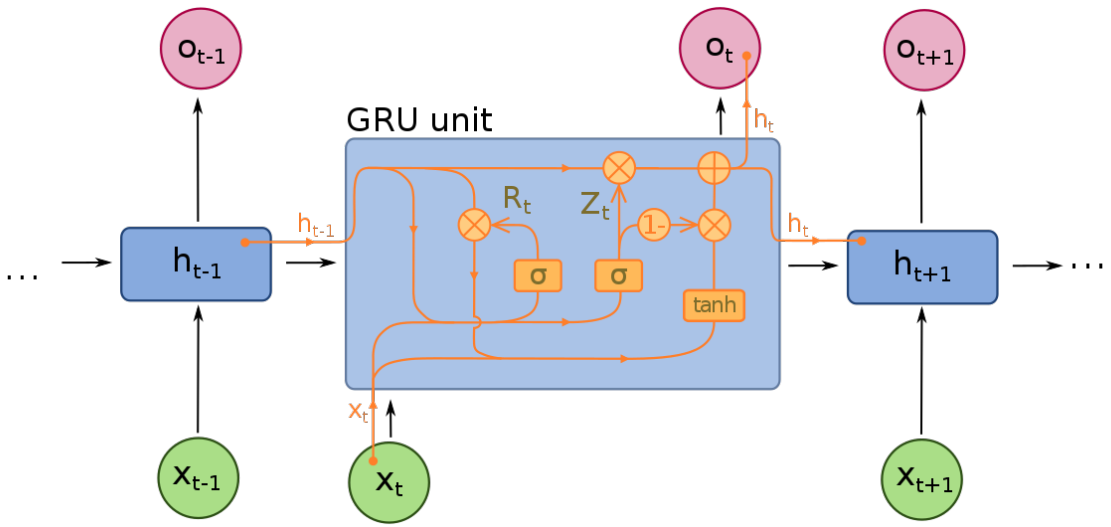
$$\boxed{h_t = o_t \odot \tanh(C_t)}$$

- **All timesteps share the same parameters:** $W_f/b_f, W_i/b_i, W_C, /b_C W_o/b_o$
- Next layers integrate h_t, not c_t
- If $f_t = 1$ and $i_t = 0 \rightarrow$ *constant error carrousel*

Images and more info can be found here.

Gated Recurrent Units – GRU

Idea : a simplfied versions of LSTM, introduced by Cho et. al. in 2014.

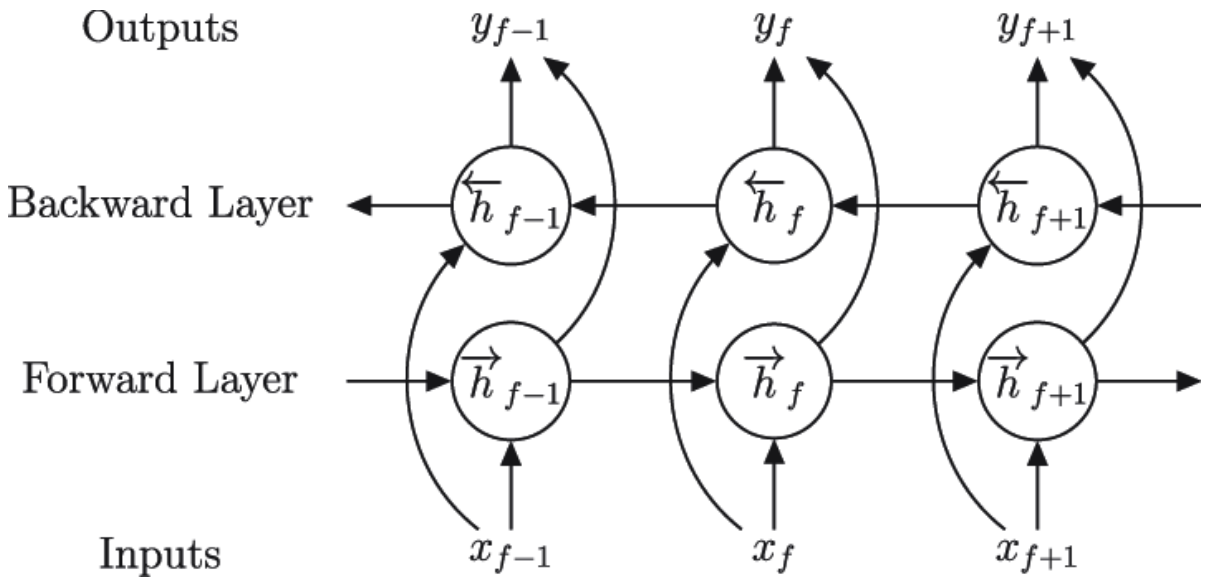


$$R_t = \sigma(W_i^x x_t + W_i^h h_{t-1} + b_i)$$
$$Z_t = \sigma(W_z^x x_t + W_z^h h_{t-1} + b_z)$$
$$n_t = \tanh(W_n^x x_t + b_{nx} + R_t \odot (W_n^h h_{t-1} + b_{nh}))$$
$$h_t = Z_t \odot h_{t-1} + (1 - Z_t) \odot n_t$$

- If $Z_t = 1 \rightarrow h_t$ is not modified
- If $Z_t = 0 \rightarrow h_t$ fully-updated

Bidirectional RNN/LSTM/GRU

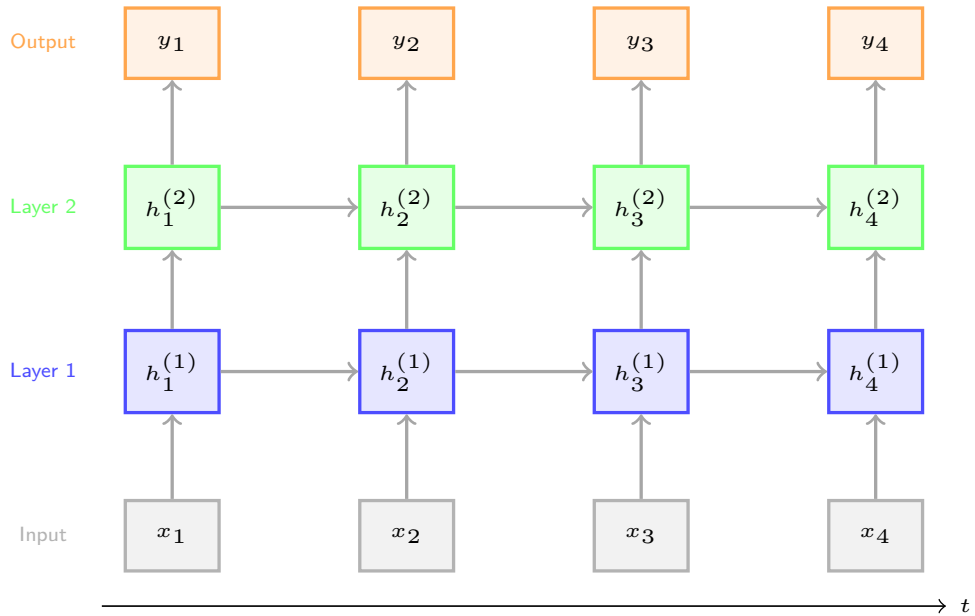
Idea : Take into account both past and future contexts to classify current sample.



- Architecture adopted when *we don't care* to build causal systems

Stacked RNNs

Idea : The hidden state of each layer becomes the **input sequence** of the layer above.



Most notable works:

- Listen, Attend ans Spell.
- Machine translation with stacked unidirectionnal LSTMs (Seq2Seq)
- Saddle point detection

Training strategies

Initialization

- LSTM: high forget bias $\rightarrow$ by default favor to remember everything
- Identity weight initialization – proposed by Le et. al.
- Orthonormal weight initialization – proposed by Henaff et. al., aims to keep: $\|Wh\|_2^2 = h^\top W^\top W h = h^\top h = \|h\|_2^2$

Training

- Layer Normalization – mean and variance are computed independently per sample over the whole layer

Regularization techniques

- gradient and activation clipping
- cannot use regular dropout – see (ZoneOut, rnnDrop)
- check this blog for more info about RNNs

Sequence Analysis

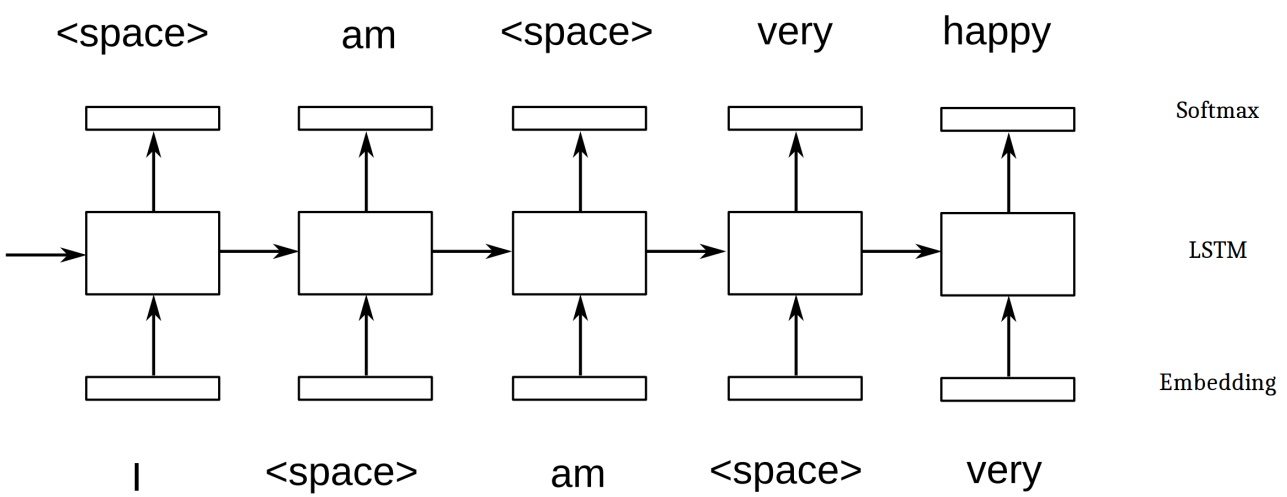
Language model using charRNN

Idea : given some fixed-length words, predict the next word/character

- Model $p(x_t|x_0, \dots x_{t-1})$ to determine the most likely word:

$$y_t = \arg\max_{y \in \mathcal{V}} p(y|x_0, \dots x_{t-1})$$

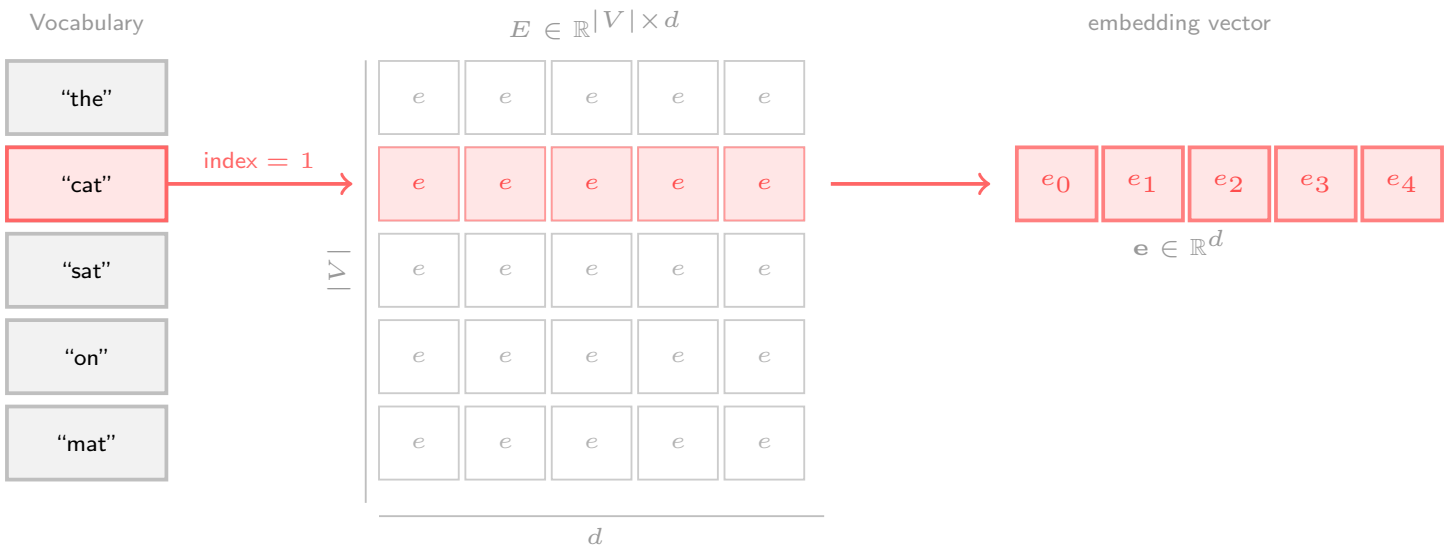
- It's a many-to-many problem during training, but many-to-one during the inference
- Check The Unreasonable Effectiveness of Recurrent Neural Networks.



The Embedding Layer

Key Idea

An embedding layer is a **lookup table** — a learned matrix $E \in \mathbb{R}^{|V| \times d}$ where each row is the dense vector representation of a vocabulary item.



Example

Example from CentraleSupelec course.

- Vocabulary : each element has a code associated.
  e.g. `'\t':0`, `'\n':1`, `'!':2`, `'.':3` ... `'A':25`, `'B':26`, ..., `'Z':51`, ...
- Dataset: windows containing 60 characters:
  e.g. Input: [4, 71, 67, 45, 36, 14, ... 100,65,36,75,89,57,70]
  e.g. Output : [71, 67, 45, 36, 14, ... 100,65,36,75,89,57,70, **63**]
- Architecture: Embedding, LSTM ($\times 2$), ReLU, Softmax
- Loss : cross-entropy **averaged over batch_size** $\times$ **seq_len**
- Optimizer: ADAM(0.01)

The training process

- To sample from a LM, you start from a context e.g. [‘L’, ‘A’, ‘G’, ‘R’, ‘E’, ‘N’, ‘O’, ‘U’, ‘I’, ‘L’, ‘L’, ‘E’ ’ ‘]
- Note that here the text is generated character by character!

Initialization sample (200 characters):

LA GRENOUILLE y
-Ô)asZYc5[h+IÉë?8—>Y.bèqp;ÎzÇÇ<)!|f]Lt+«-u
XûoÜ:!ïgVùb|Ceü9ùÈ«à
6)ZàÀçBJi)X:ZÛdzxQ8PcviV]O]xPX,Înc.è’Pâs:X;ûfjBâ?X
*ç’ED’fSOl*Z(È’È1SnjàvPïLoUÊêàDgùO9z8eJûRYJ?Yg*
Uâp|jCbû—HxBràZBMZÛPCGuR’]ÀiÊÂSBF4D),û

After 30 epchs (200 characters):

LA GRENOUILLE ET MOURE ET LA RENARDIER
Quel Grâce tout mon ambassade est pris.
L’un pourtant rare,
D’une première
Qu’à partout tout en mon nommée et quelques fleuris ;
Vous n’oserions les Fermerois, les heurs la

After 50 epochs (200 characters):

LA GRENOUILLE D’INDÉTES
[Phèdre]
Tout faire force belle, commune,
Et des arts qui, derris vôtre gouverne a rond d’une partage conclut sous besort qu’il plaît du lui dit Portune comme un Heurant enlever bien homme,

More details about this in Joel Legrand’s course.

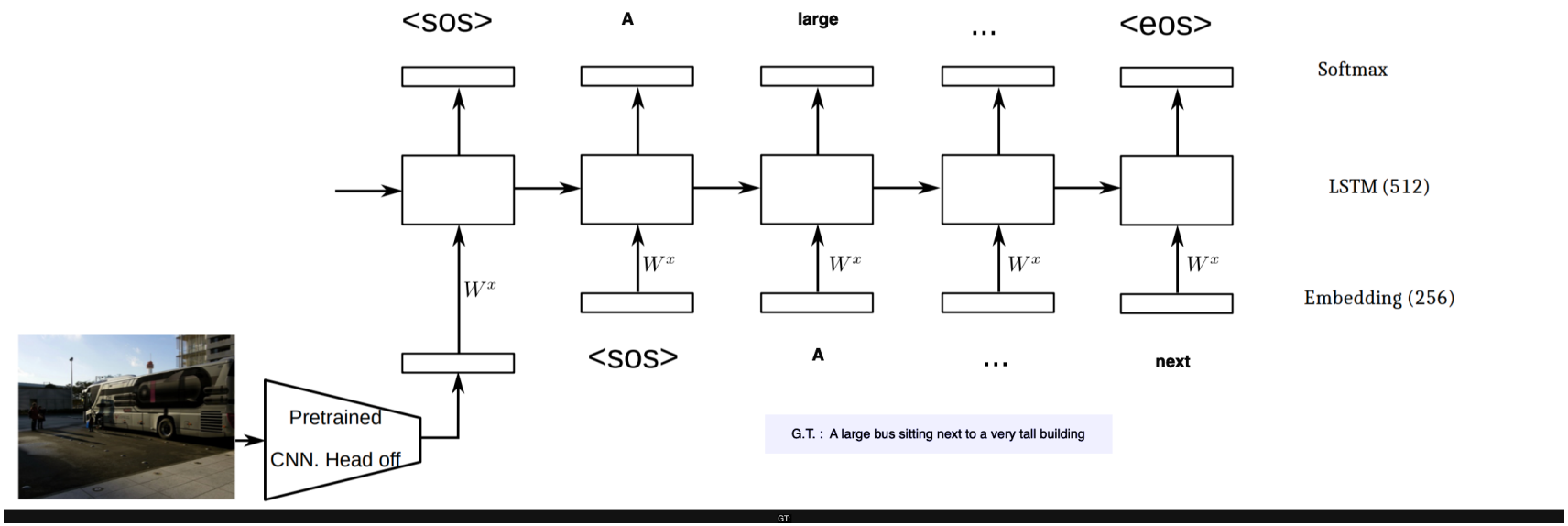
Image captioning

Idea : Based on an input image, generate a short description of it.

- One of the first works was introduced by Erhan et. al.

Show and Tell

Idea : introduce an end-to-end neural network model that combines a convolutional neural network (CNN) for image feature extraction with a recurrent neural network (RNN) for sequence generation.



Minimization problem:

$$-log(P(S_0, S_1, S_2, \dots S_T|I, \theta)) = -\sum_j log(P(S_j|S_0, \dots S_{j-1}, I, \theta))$$

Show and Tell

Training :

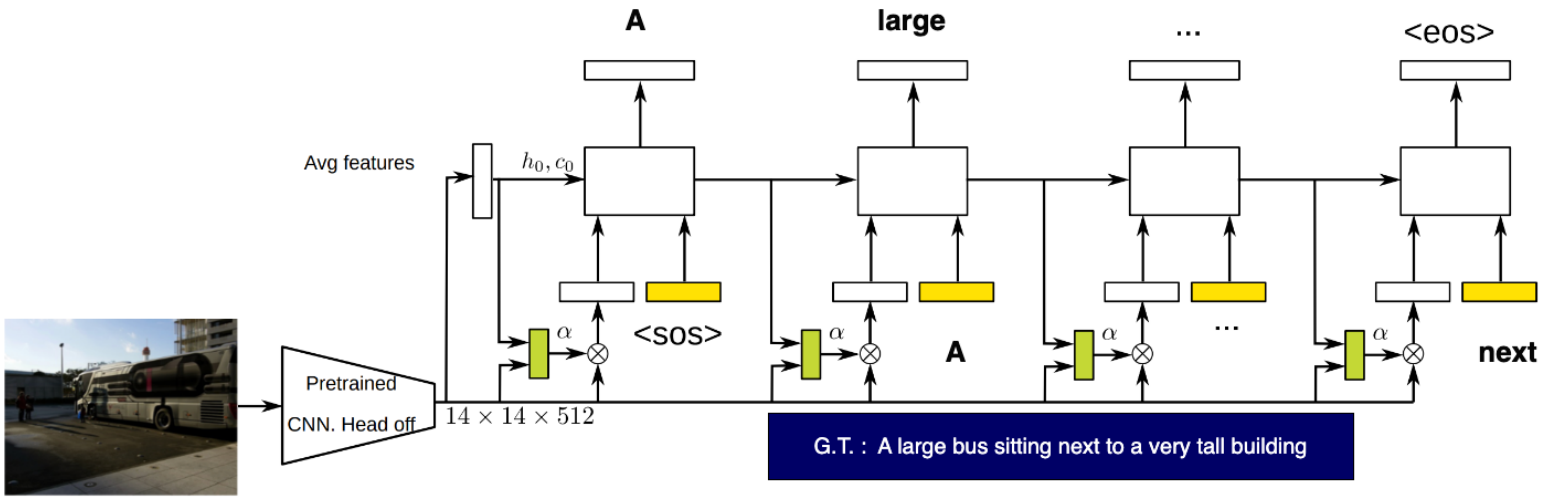
- Trained on COCO Caption dataset
- The CNN part was pretrained on Image-Net
- Words embeddings randomly initialized
- 512 - LSTM cells and embeddings
- Training the LSTM with frozen CNN then finetuning everything
- End-to-end training fails (if you start too early)
- Scheduled sampling

Inference :

- Use beam-search for decoding – more on that later

Attention based show and tell

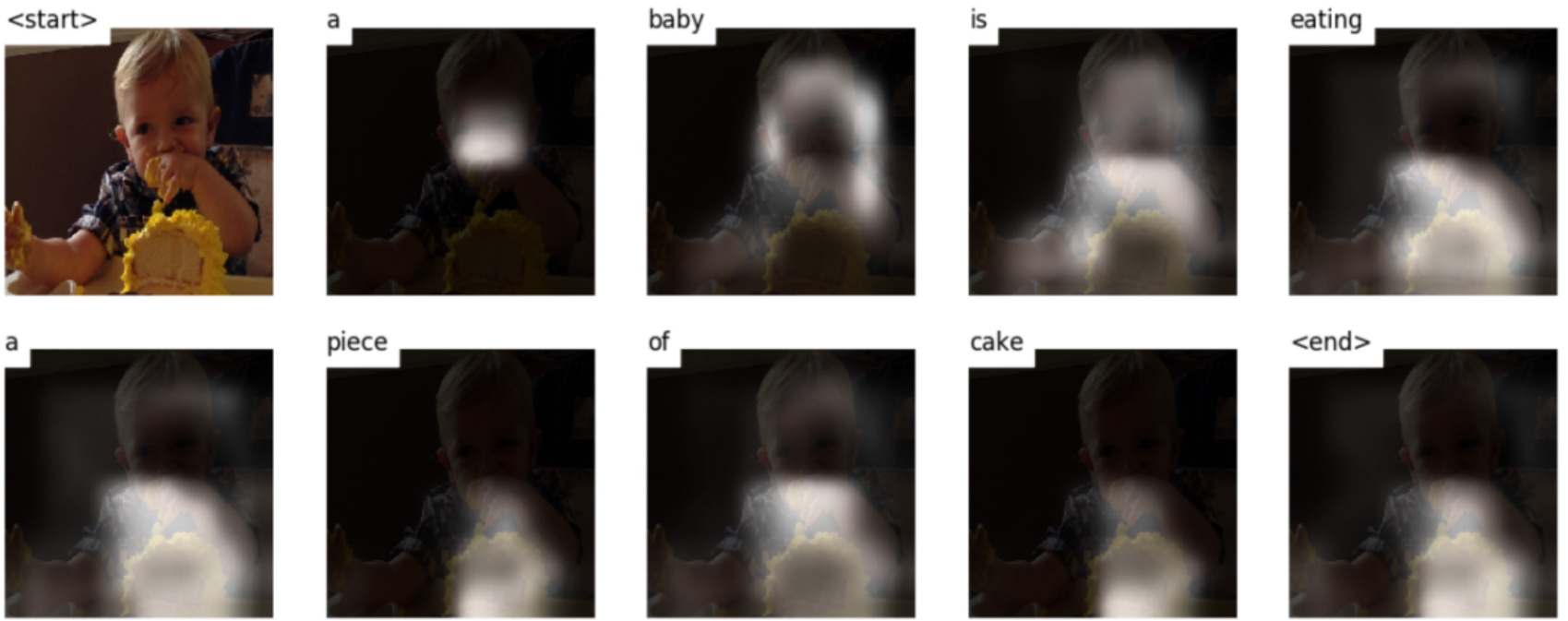
Idea : Introduces an attention mechanism into the image captioning task, allowing the model to focus on different parts of the image when generating captions.
Introduced by Xu et. al.



Stochastic Attention

- $\sum_i \alpha_{t,i} = 1 \rightarrow$ the coefficients are normalized by design
- force the model to pay attentions to all locations by using a regularization term, i.e. $\lambda \sum_l = (1 - \sum_t \alpha_{t,l})^2$, where l is the location.

Attention example



Results

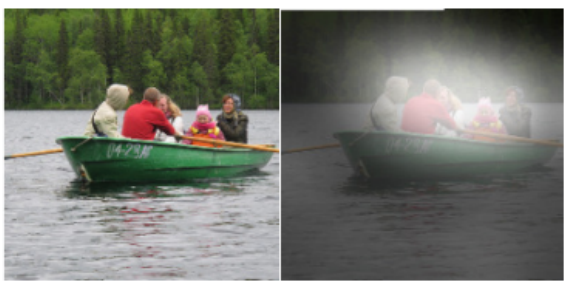
Results taken from Xu et. al.

A woman is throwing a frisbee in a park.

A dog is standing on a hardwood floor.

A stop sign is on a road with a mountain in the background.

A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.

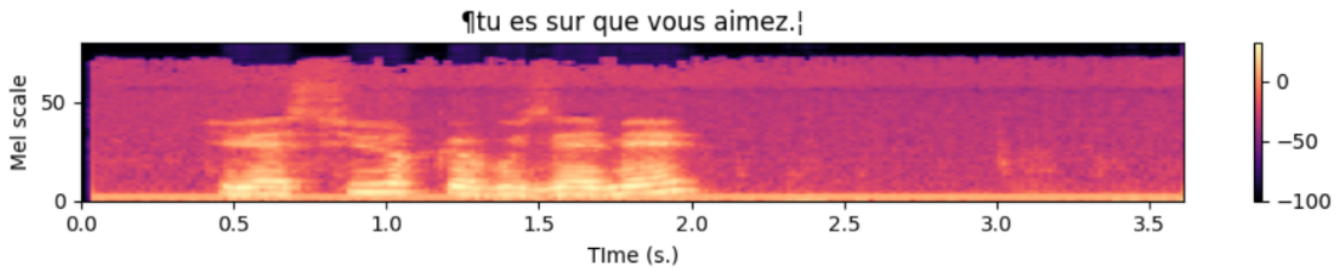
A giraffe standing in a forest with trees in the background.

Unalignment issue

Problem : How to handle Input/Output Sequences of **variable sizes** and **misaligned lengths**?

This is the case in applications such as:

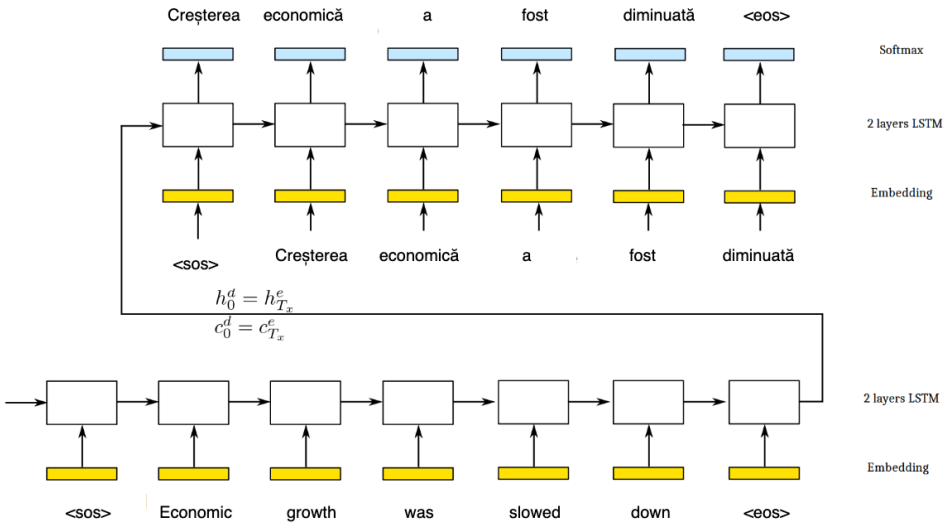
- Automatic Speech Recognition (ASR)
  - Input: Mel Spectrogram
  - Output: associated text



- Machine Translation (MT)
  - Input: *The economic growth was slowed down.*
  - Output:
    Creșterea economică a fost diminuată.
    La croissance économique a été ralentie.

Sequence to sequence using Encoder-Decoder Architecture

Idea : Encode the input sequence into a hidden state and decode the output sequence from it. Introduced in Vinyals et. al. for Neural Machine Translation.



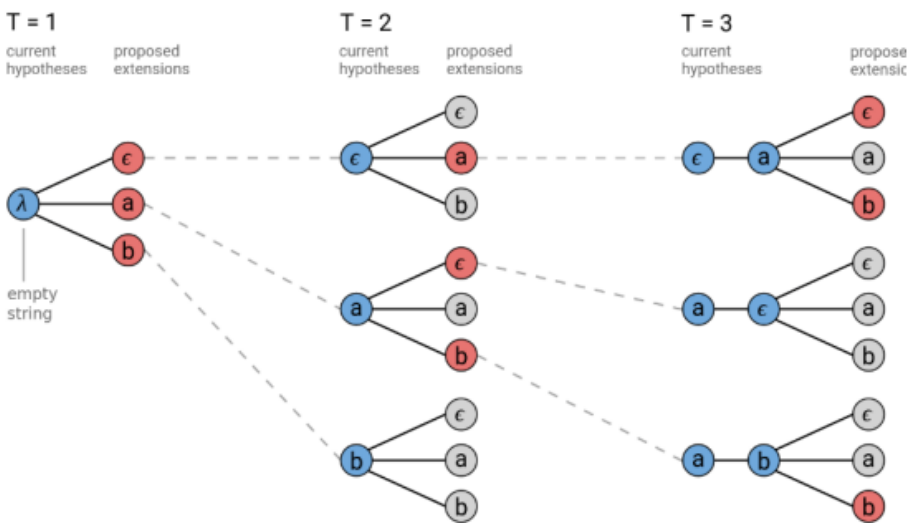
Training startegy

- Beam search decoding
- Input fed in reverse order
- Scheduled sampling
- Professor forcing
- Check this blog

Beam Search Decoding

$$p(y|x) = p(y_0|x, \theta) \prod_t p(y_t|y_0, \ldots, y_{t-1}, x, \theta)$$

- minimize $p(y|x) \rightarrow$ obtain the translation with the highest probability
- the probability distribution over the labels is dependent on the previously generated label $\rightarrow$ approximate a search by maintaining a set of M candidates
- check this for more details.



Attention in Neural Networks

Why Attention?

What if we are dealing with very long sequences?

- The RNN encoder would not be capable to compress all important information into the final hidden state
- Hence, the RNN decoder won't be able to produce meaningful outputs

Idea : look at all previous encoder and decoder outputs before predicting the next item

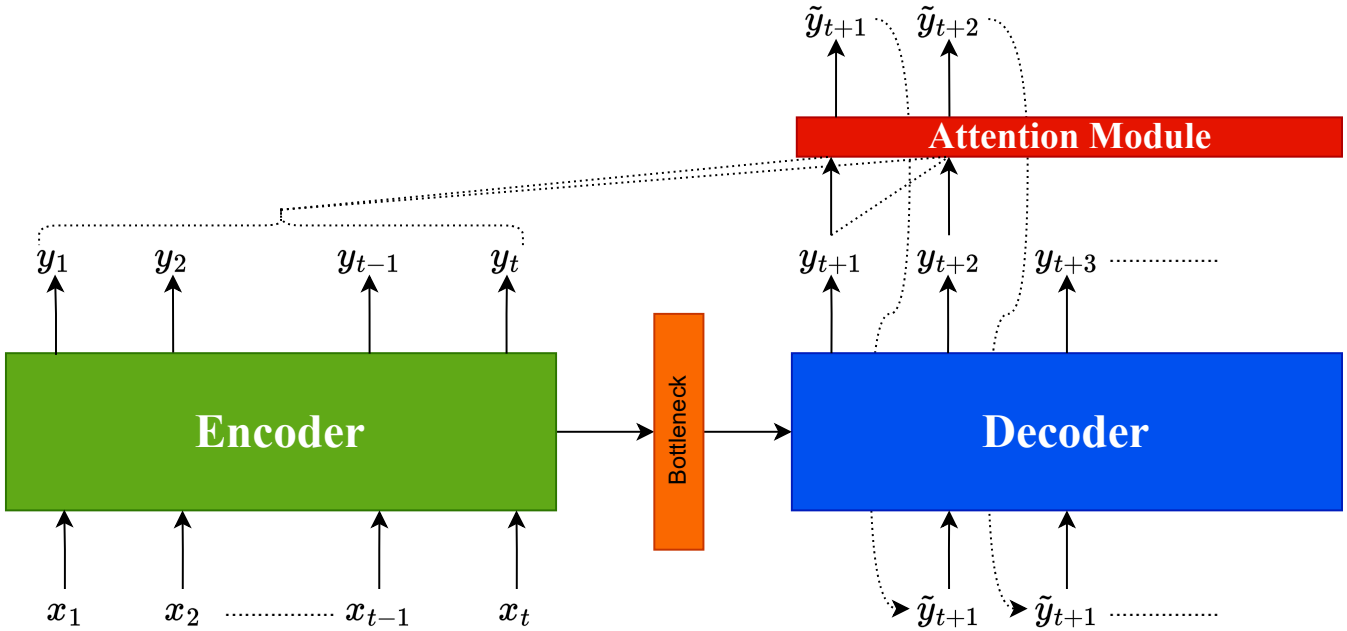


Figure: Encoder-Decoder with Attention

Self-Attention

Given a temporal sequence $\{x_i\}_{i=1}^{T}$, $T \geq 2$, $x_i \in \mathbb{R}^d$, represented in matrix form as $X \in \mathbb{R}^{T \times d}$, self-attention produces another sequence $\widetilde{X} \in \mathbb{R}^{T \times d}$ as follows:

1. Compute Keys, Queries, Values:

$$K = XW^K \qquad Q = XW^Q \qquad V = XW^V$$

$$W^K, W^Q \in \mathbb{R}^{d \times d'} \qquad W^V \in \mathbb{R}^{d \times d}$$

2. Compute similarity matrix $A \in [0,1]^{T \times T}$ with:

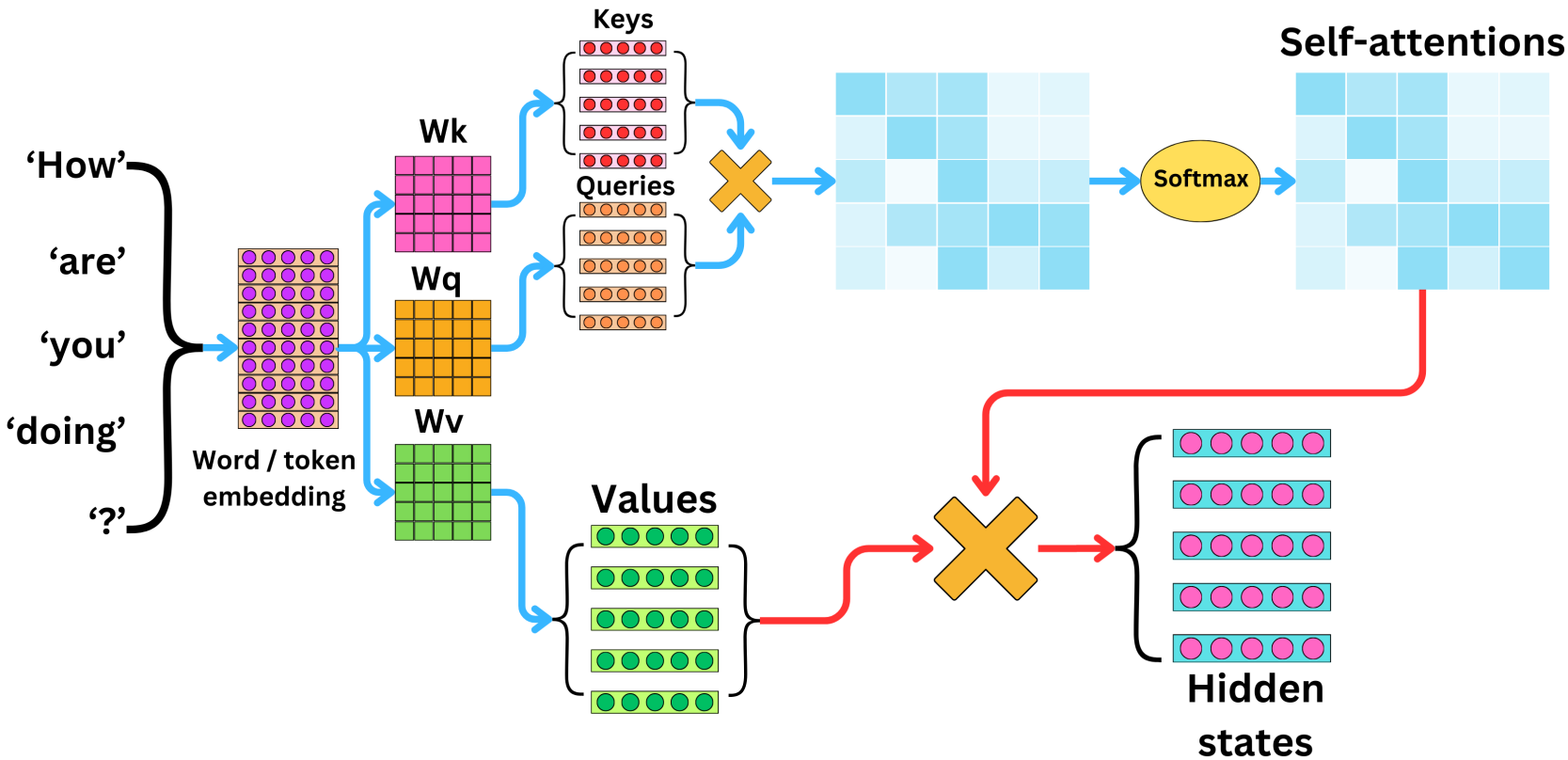
$$A_{i,:} = \text{Softmax}\left(\frac{(QK^\top)_{i,:}}{\sqrt{d'}}\right) \in \mathbb{R}^T \text{(row-wise)}, \quad \forall i \in \{1, \ldots, T\}$$

3. Compute output sequence as:

$$\widetilde{X} = AV$$

In this case, the temporal items of $\widetilde{X}$ roughly correspond to convex combinations of the initial x_i.

Self-Attention

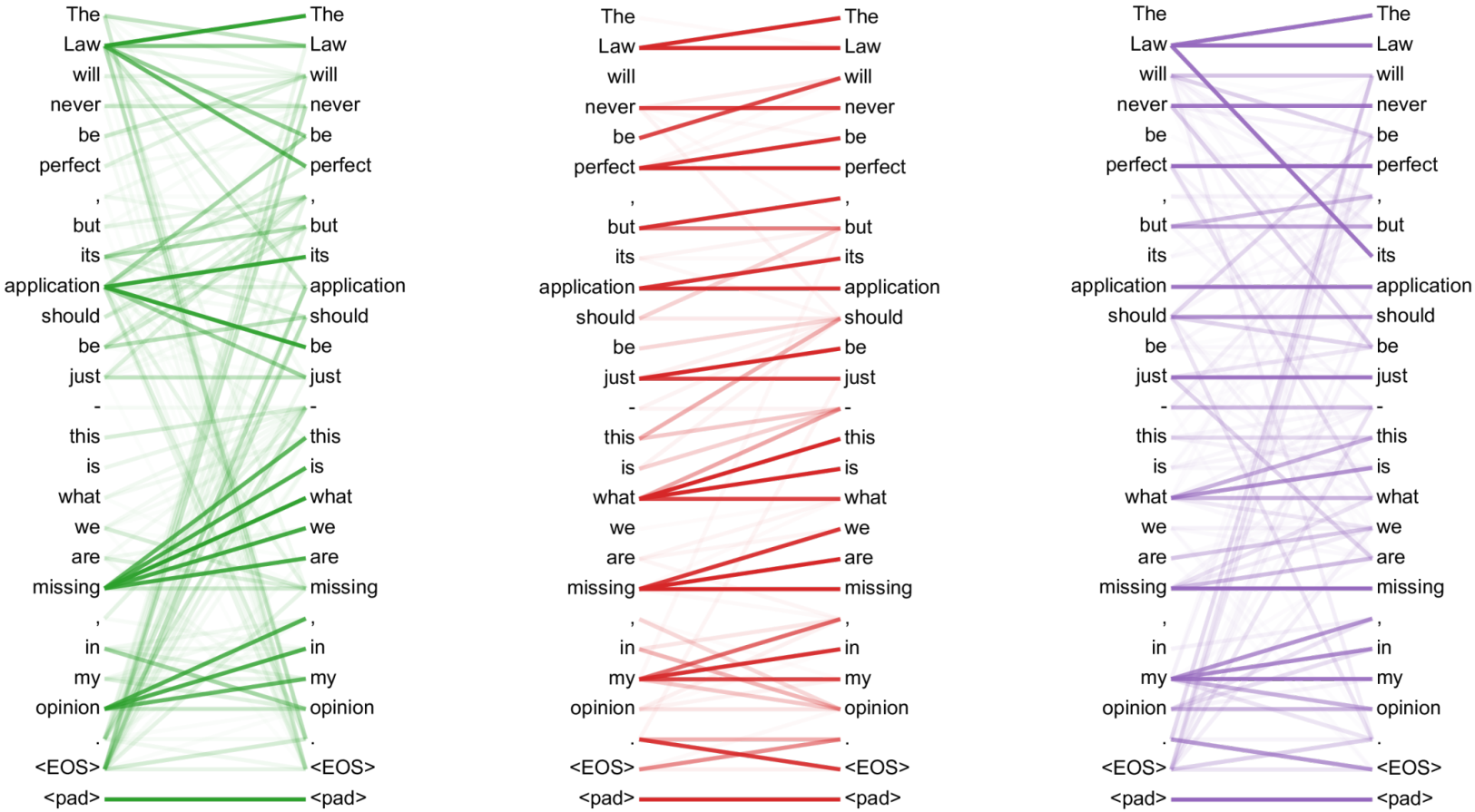


Self-attention applied directly on word embeddings. [Source]

Multi-Head Self-Attention

- Dividing by $\sqrt{d'}$ helps with stabilizing training, reducing the magnitude of attention scores
- The previous mechanism describes the inners of a *single-headed* self-attention
- For *multi-head* self-attention, with H heads:
  1. For each head $i \in \{1, \ldots, H\}$, compute triplets:
  $$(K_i, Q_i, V_i), \quad \text{using} \quad (W_i^K, W_i^Q, W_i^V)$$
  2. For each head i compute similarity matrix $A^{(i)}$, and output $\widetilde{X}_i = A^{(i)} V_i$
  3. Concatenate all $\widetilde{X}_i \in \mathbb{R}^{T \times d}$ column-wise into
  $$\widetilde{X}_c = [\widetilde{X}_1, \ldots \widetilde{X}_H] \in \mathbb{R}^{T \times (d \cdot H)}$$
  4. Compute final output:
  $$\widetilde{X} = \widetilde{X}_c W^O \in \mathbb{R}^{T \times d}, \quad \text{with} \quad W^O \in \mathbb{R}^{(d \cdot H) \times d}$$

Self-Attention



Example of similarities learned by different self-attention heads, in the same layer. Illustrations from Attention is All you Need

Cross-Attention

What if we want to represent one sequence in terms of another?

Given a sequence $Y \in \mathbb{R}^{T_Y \times d}$ which we want to represent in terms of sequence $X \in \mathbb{R}^{T_X \times d}$, $T_Y, T_X \geq 2$, cross-attention works as follows:

1. K_i, V_i are computed using X, Q_i is computed using Y
2. $A^{(i)}$ will now have size (T_Y, T_X)
3. Output is computed similarly as:

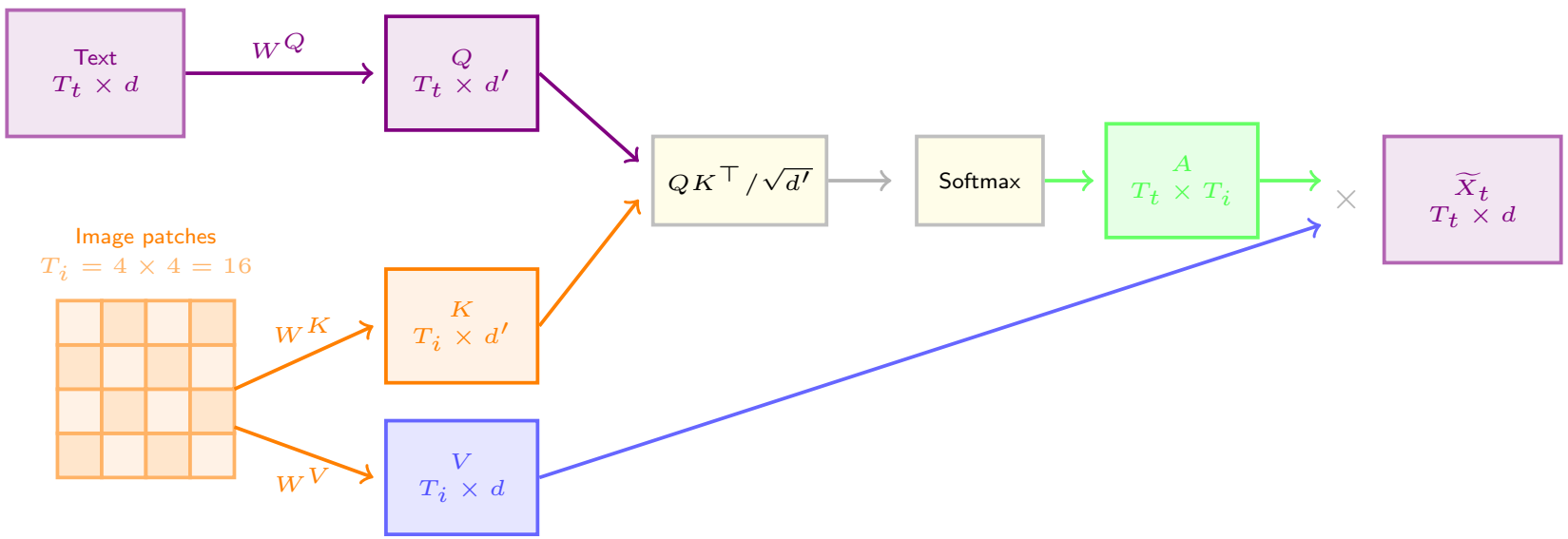
$$\widetilde{Y}_i = A^{(i)} V_i = A^{(i)} X W_i^V$$

Therefore, each element of $\widetilde{Y}_i$ will be represented as a convex combination of linearly projected elements of X.
For each element $\widetilde{y}_i$, one can use $A^{(i)}$ to retrive the most similar $\widetilde{x}_i$.

Cross-Attention Applications — Text and Image

Key Idea

Queries come from the **text**, while Keys and Values come from the **image** — each text token attends over the image features.



$A \in \mathbb{R}^{T_t \times T_i}$ — each text token attends over all image patches.

Related applications: image captioning, Vision Question Answering (VQA)

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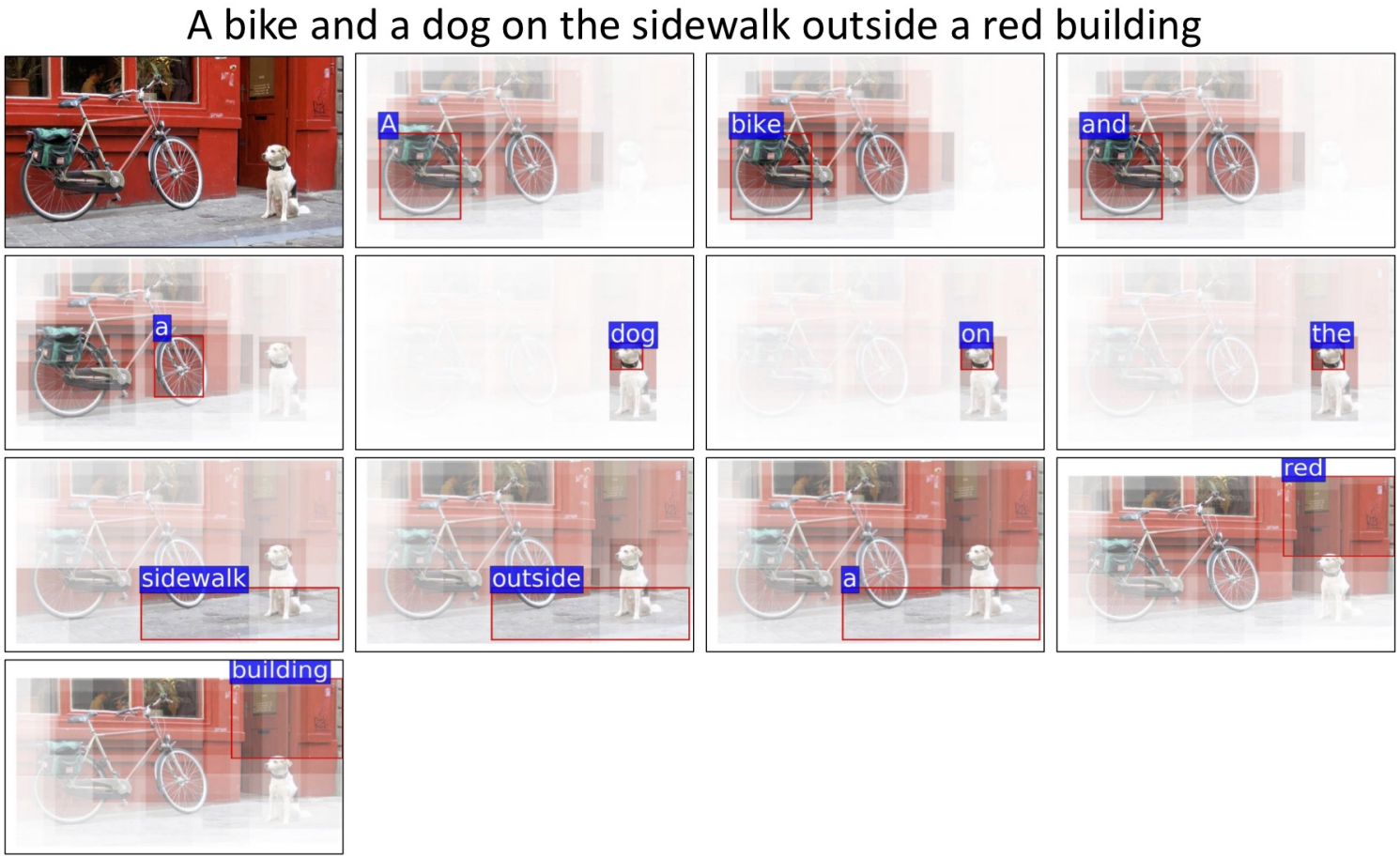

Alignment learned between a sequence of words and image regions (represented as a sequence of patches). Example from Stacked Cross Attention for Image-Text Matching

Cross-Attention Applications — ASR (Text and Audio)

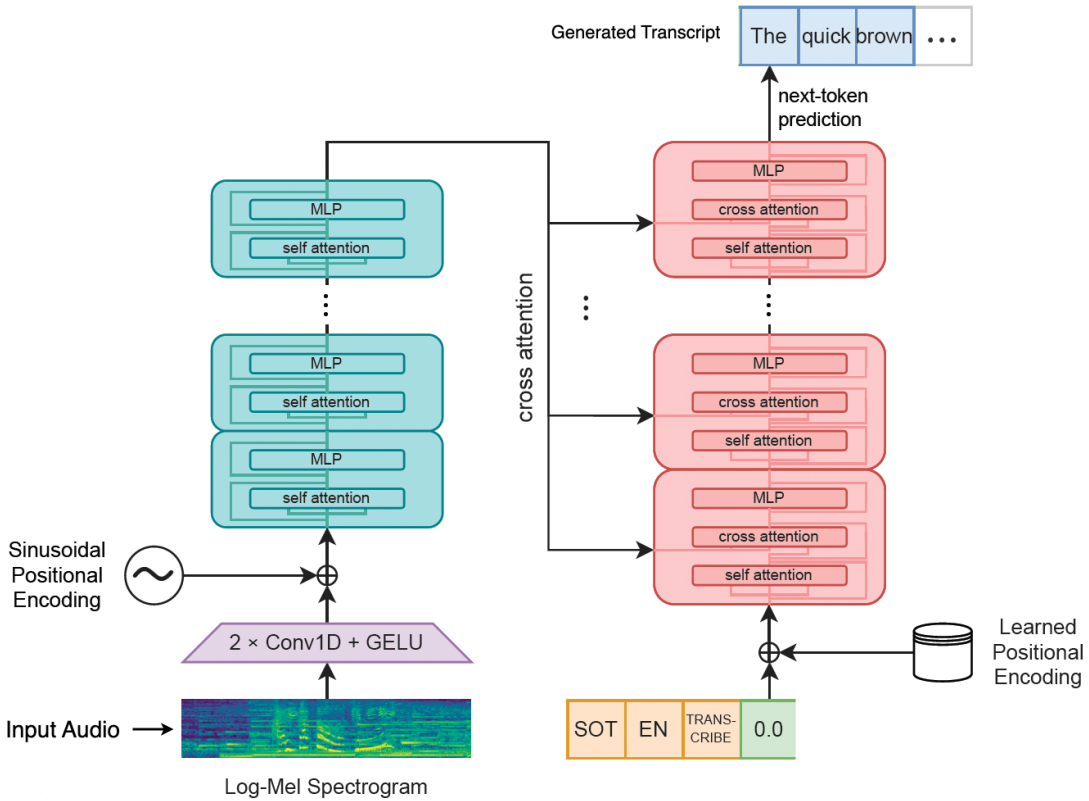

Figure: Architecture of `Whisper` ASR model. Source

- Encoder processes the input audio
- Text-decoder is modulated by audio features through cross-attention modules, generating the corresponding transcript